Classical Machine Learning for Aviation Engine Health Monitoring: Anomaly Detection and Fault Classification on the NGAFID Benchmark

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`https://github.com/Barbosa12/NGAFID`

***Abstract*—General aviation still relies on calendar-based maintenance, which misses faults that emerge between fixed inspection windows. We study whether *classical* machine learning on flight-level statistical features can support condition-based maintenance, using the National General Aviation Flight Information Database (NGAFID), a real-world benchmark of $11{,}446$ Cessna-172 flights labeled with 19 fault categories. We cast the problem as a two-stage pipeline: binary Anomaly Detection (AD) followed by 19-class Fault Classification (FC). Five classical models are trained over a hierarchy of representations—global per-sensor statistics, temporal segments, and physics-based inter-cylinder features—and compared with deep learning (DL) figures reported in the literature. Our best classical system reaches $F1 = 0.703$ on AD (tuned ensemble, 224 features), within 1.8 points of a published temporal DL model, but FC tops out at $F1 = 0.207$ for a flat classifier and $F1 = 0.459$ for an Oracle hierarchical ceiling. We trace this asymmetry to *target-component sparsity*: fault-discriminative information lives in low-variance principal components (PC9, PC19–23) that global statistics discard. An end-to-end error decomposition shows that fault *typing*, not detection, is the bottleneck (the largest single error mode). We report the DL comparison transparently: its source preprint was later withdrawn by its authors over metric-computation flaws, so those numbers are treated as indicative only. All results derive from the project notebooks 01–05.**

***Index Terms*—anomaly detection, fault classification, time series, aviation maintenance, NGAFID, imbalanced learning.**

I. Introduction

Modern aircraft continuously log dozens of sensor channels through their engine control units, yet general aviation maintenance remains *calendar-based*: inspections occur at fixed intervals regardless of component condition. This reactive paradigm misses faults that emerge between windows, creating safety risks and costly unscheduled groundings. Machine learning offers a path toward *condition-based* maintenance, flagging anomalous flights and identifying likely fault types directly from telemetry.

We use the NGAFID benchmark [1], a real (non-simulated) dataset from an active Cessna-172 flight school, and address two research questions: **(RQ1)** can a classical classifier trained on statistics extracted from the full flight profile reliably separate healthy from anomalous flights? and **(RQ2)** given an anomalous flight, can the same approach identify which of 19 fault types is responsible? We hypothesize that *global* statistics suffice for AD (whole-flight health state), but fail on FC because fault signatures are *local* micro-patterns that global aggregation discards. Both tasks are composed into an end-to-end (E2E) 20-class pipeline.

Our contributions are: (i) a systematic evaluation of five classical models over a hierarchy of feature representations on NGAFID; (ii) empirical evidence for *target-component sparsity* as the structural reason FC is hard; (iii) an Oracle hierarchical experiment that raises the classical FC ceiling from 0.116 to 0.459; and (iv) an E2E error decomposition that locates fault classification as the dominant bottleneck. Every figure and table in this paper is reproduced from the project notebooks 01 (EDA), 02 (dataset/feature build), 03 (AD), 04 (FC), and 05 (E2E).

II. Dataset and Feature Engineering

A. NGAFID Benchmark

NGAFID correlates 23-channel, 1 Hz telemetry with verified unscheduled maintenance records [1]. We use the 2-day benchmark subset: $11{,}446$ flights, nearly balanced at the binary level—$5{,}844$ healthy (51.1%) and $5{,}602$ anomalous (48.9%)—but *severely long-tailed* at the fault level, where the two largest categories (rocker-cover and intake- gasket damage) dominate and the smallest classes have barely a hundred flights, an imbalance ratio of roughly 38:1 (Fig. 1). Each flight is interpolated to a fixed $\mathbf{X} \in \mathbb{R}^{2048\times 23}$ matrix via cubic spline. The 23 channels span electrical (`volt`, `amp`), fuel, engine (`FFlow`, `OilT`, `OilP`, `RPM`), four-cylinder CHT and EGT, flight state, and ambient temperature. All experiments use 5-fold stratified cross-validation with strict per-aircraft isolation to prevent leakage; standardization is fit on the training split only.

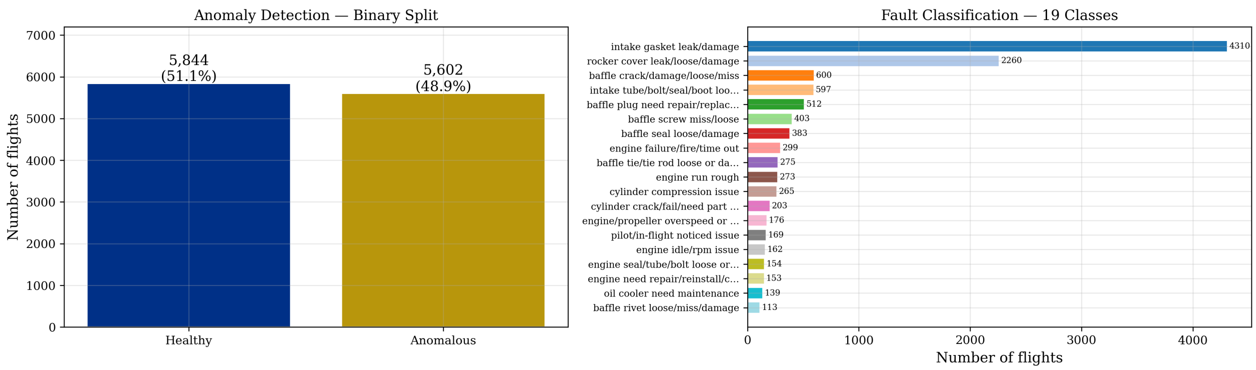

Fig. 1. NGAFID benchmark subset (notebook 01). Left: the binary AD split is nearly balanced. Right: the 19 fault classes are severely long-tailed, with the two head classes dominating—the core difficulty for fault classification.

TABLE I
FEATURE REPRESENTATIONS EXPLORED (NOTEBOOK 02).

Repr.	What it captures	Dim.
B	Global per-sensor statistics	184
B_{seg}	Repr. B over N flight segments	552–1840
F_{++}	B + cross-cylinder asymmetry/ratios	224
F	F_{++} + XGB gain selection	60

B. *Feature Representations*

We compress each variable-length flight into a fixed feature vector (Table I). Representation B computes eight statistics $(\mu, \sigma, \min, \max, \text{range}, \text{skew}, \text{kurt}, \hat{\beta})$ per channel, giving $23 \times 8 = 184$ features. Representation F_{++} adds *cross-physical* inter-cylinder features (CHT/EGT spread, coefficient of variation, pairwise ratios), reaching 224 features; XGBoost gain-based selection then reduces this to 60 (Repr. F). Segmented variants (B_{seg}) apply Repr. B to N equal flight segments to inject coarse temporal ordering.

C. *Why Fault Classification is Hard*

Projecting Repr. B into principal-component space exposes the structural obstacle (Fig. 2). The high-variance components PC1–PC2 capture operational variability common to healthy and anomalous flights (altitude, throttle, ambient temperature) and are nearly *non-discriminative*. The components that actually separate healthy from anomalous—PC9 and PC19–PC23 by F-statistic—each contribute very little to total variance. Fault-relevant information is thus *sparse* and buried beneath operational variance, so any global statistic that emphasizes high-variance directions discards exactly the signal FC needs. This is the empirical basis for our hypothesis.

III. METHODOLOGY

The 20-class problem (0 =healthy, 1–19 =faults) is decomposed into a two-stage cascade (Fig. 3). Stage 1 (AD) is binary; only flights flagged anomalous proceed to Stage 2 (FC). The decomposition reflects the opposing inductive biases of the two tasks: AD benefits from global whole-flight context, while FC requires local fault resolution [2].

Models. We evaluate Logistic Regression (LR), Linear SVM, XGBoost (XGB) [3], Random Forest (RF), and KNN, all in scikit-learn [4], with class weighting for imbalance; a soft-voting *Ensemble* (LR+XGB) is also tuned. **Oracle hierarchy.** Ward clustering groups the 19 classes into $K{=}4$ physically coherent super-groups; a per-group XGB is trained, using ground-truth group membership at test time to obtain an upper bound. **Metrics.** The primary metric is macro-$F1$; for AD we also report AUC and the False Negative Rate (FNR), the safety-critical miss rate. **DL reference.** We compare against ConvTokMHSA, MMK Net, and the LMSD framework as reported in the literature.[1]

IV. RESULTS

A. *Anomaly Detection*

Table II reports 5-fold AD performance. On Repr. B the tuned Ensemble leads at $F1 = 0.691$; adding cross-cylinder features (F_{++}) raises the best classical result to $F1 = 0.703$ with AUC $= 0.767$ and FNR $= 0.307$ (Fig. 5). This is *competitive* with temporal DL: it trails the best reported model (ConvTokMHSA, 0.799) by 9.6 points but is within 1.8 points of MMK Net (0.721). Feature importance (Fig. 4) is dominated by oil-pressure statistics (`E1 OilP_std`, `_range`, `_mean`), matching the physical role of oil pressure as the primary engine-health indicator, followed by electrical and cross-cylinder features.

B. *Fault Classification*

FC is markedly harder (Table III). On the anomalous flights, flat XGBoost on Repr. B reaches only $F1 = 0.116$ (default) and 0.176 (tuned); the best flat classical result, XGBoost on F_{++}, attains $F1 = 0.207$. Per-class analysis (Fig. 6) shows the long-tail effect directly: only the two head classes clear $F1 > 0.30$, most classes land in a 0.10–0.30 "partial" band, and per-class $F1$ correlates with training-set size—a textbook

[1]The deep learning figures are reproduced from the LMSD preprint [2], which was subsequently *withdrawn* by its authors, who stated that "significant methodological flaws have been identified in the experimental validation and metric-computation procedures." We present these numbers as indicative reference points, not validated upper bounds, and interpret all gaps accordingly.

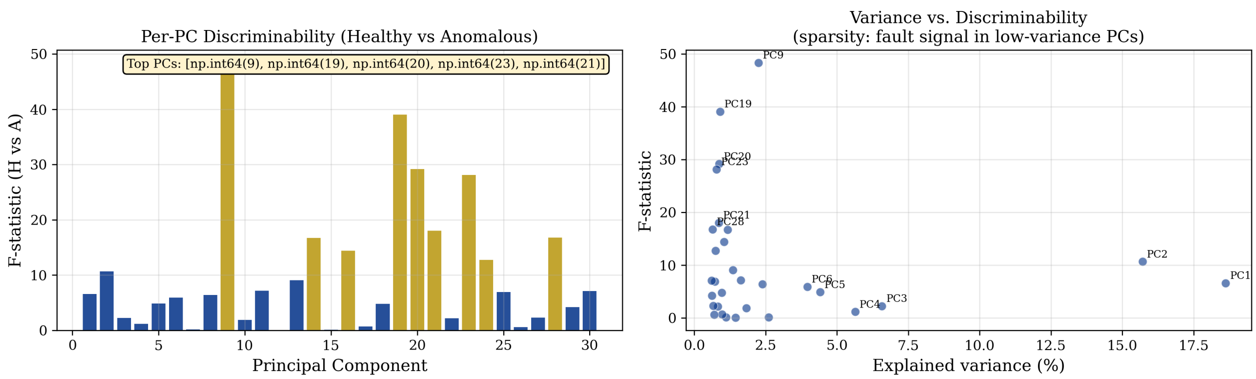

Fig. 2. Target-component sparsity (notebook 01). Left: per-PC discriminability (F-statistic, healthy vs. anomalous)—the most discriminative components are high-index. Right: discriminability vs. explained variance—the high-variance PC1–PC2 are weakly discriminative, while the discriminative PCs carry little variance.

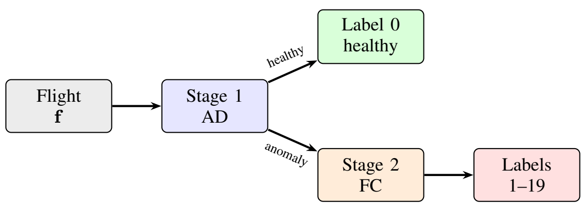

Fig. 3. Two-stage diagnosis pipeline (notebook 05).

TABLE II
ANOMALY DETECTION (5-FOLD CV). BEST CLASSICAL PER METRIC IN **BOLD**; DL FIGURES ARE INDICATIVE (FOOTNOTE 1).

Model	**Repr.**	**F1**	**AUC**	**FNR**	**FPR**
Classical (this work, Repr. B)					
Ensemble	B	0.691	0.754	0.322	0.296
Logistic Reg.	B	0.679	0.730	0.308	0.333
XGBoost	B	0.675	0.742	0.330	0.321
Linear SVM	B	0.675	0.726	0.347	0.303
Random Forest	B	0.651	0.707	0.400	0.298
KNN	B	0.545	0.562	0.477	0.431
Ensemble (best)	F_{++}	**0.703**	**0.767**	**0.307**	**0.287**
DL benchmarks (indicative) [2]					
ConvTokMHSA	raw	0.799	—	0.163	—
InceptionTime [5]	raw	0.784	—	0.196	—
MMK Net	raw	0.721	—	0.298	—

imbalance signature. Hierarchical grouping helps: an Oracle $K{=}4$ partition lifts the ceiling to $F1 = 0.459$ (Fig. 7), but the realistic cascade variant (0.172) confirms that the gain depends on knowing the coarse fault group. Even the Oracle ceiling trails the best reported DL model (MMK Net, 0.623) by 16 points; tellingly, global-attention ConvTokMHSA (0.208) barely matches our flat XGBoost, confirming that whole-flight context does not help fault typing.

C. End-to-End Diagnosis

Composing both stages into a 20-class decision (Table IV), the cascade AD→FC reaches $F1 = 0.131$, beating a direct 20-class classifier (0.113) and recovering AD recall of 0.688. The error decomposition (Fig. 8) is the key diagnostic: of all cascade predictions, 50% are correct, 15.3% are Stage-1 false negatives (faulty flights labeled healthy—the safety-critical mode), 14.7% are Stage-1 false positives, and 20.0% are Stage-2 wrong-type errors. **Fault typing (S2) is thus the single largest error source**, confirming that FC—not detection—is where the pipeline must improve.

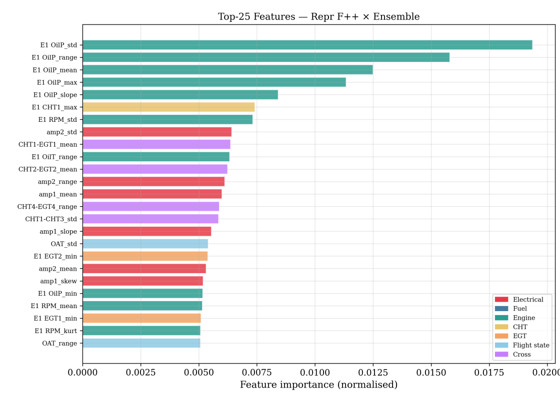

Fig. 4. AD feature importance, tuned Ensemble on F_{++} (notebook 03). Oil-pressure statistics dominate; colors denote sensor groups.

TABLE III
FAULT CLASSIFICATION (5-FOLD CV, MACRO-$F1$, NOTEBOOK 04).

Model	**Repr.**	**F1 macro**
XGBoost (default)	B	0.116
XGBoost (tuned)	B	0.176
XGBoost (tuned, best flat)	F_{++}	**0.207**
XGB cascade $K{=}4$	F_{++}	0.172
Oracle $K{=}4$ (ceiling)	F_{++}	**0.459**
MMK Net (indicative) [2]	raw	0.623
ConvTokMHSA (indicative)	raw	0.208

V. DISCUSSION

Answering the research questions. For **RQ1**, the answer is yes: a tuned classical ensemble on global statistics detects anomalies competitively ($F1 = 0.703$, within 1.8 points of MMK Net), is interpretable, and trains on a laptop—enough

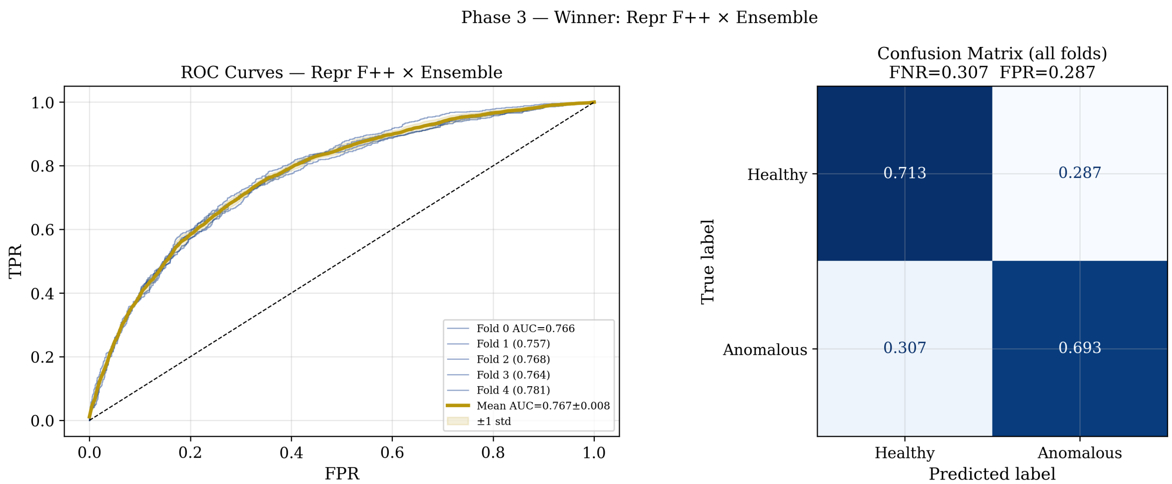

Fig. 5. AD winner—Ensemble on F_{++} (notebook 03). Left: per-fold ROC (mean AUC $= 0.767 \pm 0.008$). Right: confusion matrix (FNR $= 0.307$, FPR $= 0.287$); 69.3% of anomalous flights are caught.

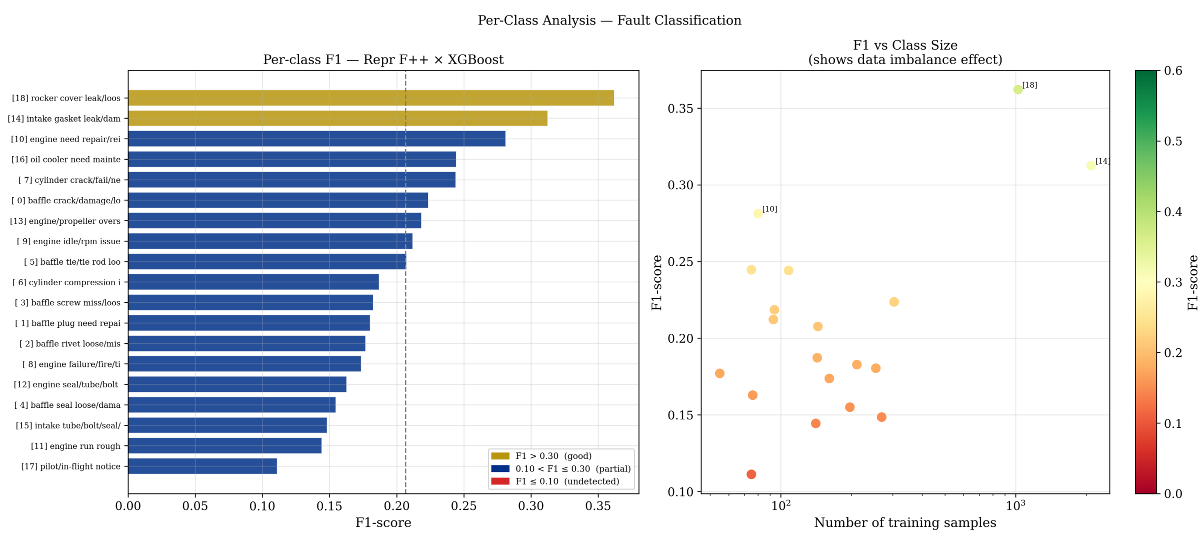

Fig. 6. Per-class FC, best flat model XGBoost on F_{++} (notebook 04). Left: per-class $F1$; only head classes exceed 0.30. Right: $F1$ vs. training-set size, exposing the imbalance effect.

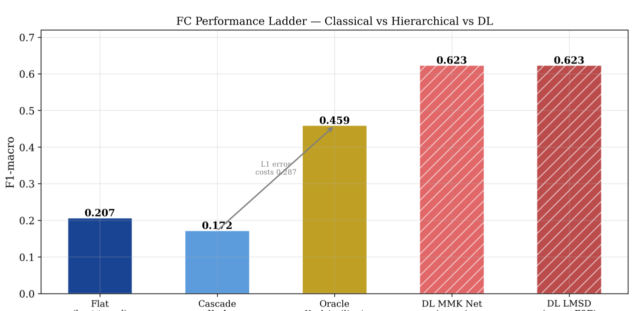

Fig. 7. FC performance ladder (notebook 04): flat $\rightarrow$ realistic cascade $\rightarrow$ Oracle ceiling $\rightarrow$ indicative DL.

TABLE IV
END-TO-END DIAGNOSIS (MACRO-$F1$ OVER 20 CLASSES, NOTEBOOK 05).

Architecture	**Repr.**	**F1$_{20}$**	**AD recall**
Direct 20-class	F_{++}	0.113	0.509
Cascade AD→FC	F_{++}	**0.131**	0.688
LMSD (DL, indicative) [2]	raw	0.623	0.837

to move part of a fleet toward condition-based inspection. For **RQ2**, the answer is largely no: even with feature engineering and an Oracle hierarchy, FC tops out at $F1 = 0.459$, leaving most minority faults only partially recognized. The system is fit for *triage* (sick vs. healthy) but not yet for *prescription* (which part to fix). Both outcomes match our hypothesis: the difference is explained by target-component sparsity (Fig. 2)—global statistics retain the whole-flight health state needed for AD but discard the low-variance local signatures needed for

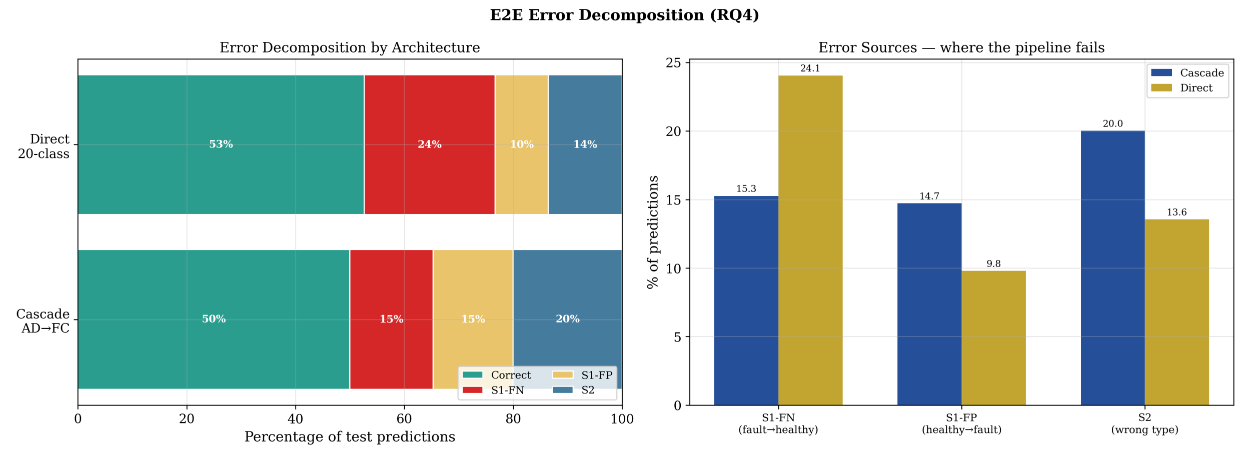

Fig. 8. E2E error decomposition (notebook 05). Left: error budget by architecture. Right: error sources; for the cascade, Stage-2 wrong-type errors (20%) are the largest, identifying fault classification as the bottleneck.

FC.

Relation to prior work. The qualitative pattern—global context helps AD, local micro-patterns help FC—is corroborated by the convolutional- transformer line of NGAFID research [6] and by our own ablation, in which global-attention ConvTokMHSA fails to beat flat XGBoost on FC. The specific DL gap magnitudes, however, derive from a preprint later withdrawn over metric-computation flaws [2]; we therefore treat them as provisional. A retraction at the state of the art arguably makes a transparent, reproducible classical baseline with honest cross-validated variance more valuable, not less.

Tools surveyed. The project used the open-source Python stack: NumPy/pandas for munging, scikit-learn [4] for the linear and distance models, pipelines, and fold-safe preprocessing; XGBoost [3] as the best flat classical model and gain-based feature selector; and Matplotlib/Seaborn for EDA and figures. For FC, even the most informative tabular representation (F_{++}) plateaus below the deep-learning reference, because global and segmented statistics discard the local temporal signatures that distinguish fault types—pointing toward locally restricted deep learning (Keras/TensorFlow) as the principled next tool, which we surveyed but left to future work.

Limitations and future work. Results use the 19-class benchmark subset; the full 36-class NGAFID is harder. The Oracle $K{=}4$ figure is an upper bound, not a deployable number, and the DL reference points inherit the reliability problem of the withdrawn source. The error decomposition makes the priority unambiguous: fault typing is the bottleneck. The clearest next steps are (i) a locally restricted temporal CNN trained on raw sequences to test whether the FC gap is genuinely temporal; (ii) re-running the benchmark once corrected DL results are published; and (iii) tackling the 38:1 imbalance directly with resampling or cost-sensitive losses.

VI. Conclusion

Classical machine learning on global flight statistics delivers a deployable, interpretable anomaly screen for general-aviation condition-based maintenance today ($F1 = 0.703$, within 1.8 points of a published temporal model), but reliable fault attribution remains the unsolved half: the classical ceiling is $F1 = 0.459$, and the end-to-end bottleneck is fault typing. Because fault-discriminative information is sparse and low-variance, invisible to global statistics, the indicated next step is a locally restricted temporal deep model, evaluated against a re-validated benchmark. All code and notebooks are available at `https://github.com/Barbosa12/NGAFID`.

References

[1] H. Yang and T. Desell, “A large-scale annotated multivariate time series aviation maintenance dataset from the NGAFID,” 2022.

[2] X. Chen, Z. Wei, Y. Hu, Z. Zeng, K. Zeng, and W. Wang, “A heterogeneous long-micro scale cascading architecture for general aviation health management,” 2026, preprint subsequently withdrawn by the authors, citing methodological flaws in the experimental validation and metric-computation procedures; reported figures are treated as indicative rather than validated.

[3] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD*, 2016, pp. 785–794.

[4] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[5] H. I. Fawaz *et al.*, “InceptionTime: Finding AlexNet for time series classification,” *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1936–1962, 2020.

[6] H. Yang, A. LaBella, and T. Desell, “Predictive maintenance for general aviation using convolutional transformers,” in *Proceedings of the AAAI Conference on Artificial Intelligence (IAAI-22)*, 2022.